Knowledge Map
What I Know, What's Known, and What's Open

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Abstract

A consolidated reference covering everything explored in this research programme: what I understand, what is known (and by whom), what I proved that turned out to be classical, what remains genuinely open, and where the understanding might lead. Not a paper. A map for finding the next problem.

Contents

1 The central question

When does observationally coherent data determine a genuine probability measure, and what additional commitments does this require?

This decomposes into three layers:

1. **Extension:** compatible finitely additive charges on a directed system of Boolean algebras $\to$ σ-additive probability measure.
2. **Realization:** measure on the dual space (Stone space) $\to$ measure on an underlying state space Ω.
3. **Value-definiteness:** probabilistic description $\to$ simultaneous definite values for all observables.

Each layer requires a commitment the previous does not. The algebraic structure of the observation algebra constrains which commitments are available; distributivity is sufficient to make all three available.

2 Layer 1: From charges to measures

2.1 The classical results (all known)

- **Continuity from above:** A finitely additive charge on an algebra is σ-additive iff $E_n \downarrow \varnothing \Rightarrow \ell(E_n) \to 0$. *Halmos, Measure Theory (1950), §13, Theorem D.*
- **Carathéodory extension:** A σ-additive premeasure on a semiring extends uniquely to a measure on the generated σ-algebra. *Carathéodory (1914); Halmos §13; Bogachev Vol. 1, Ch. 1.*
- **Kolmogorov extension:** Compatible finite-dimensional distributions on a product space extend to a unique probability measure, under standard Borel / tightness conditions. *Kolmogorov, Grundbegriffe (1933); Billingsley, Probability and Measure, Ch. 36.*
- **Choksi inverse limits:** Compatible measures on an inverse system of compact Hausdorff spaces extend to the inverse limit. *Choksi, Inverse limits of measure spaces, Proc. London Math. Soc. (1958), Theorem 1.*
- **Yosida–Hewitt decomposition:** Every bounded finitely additive charge decomposes uniquely as $\ell = \ell_c + \ell_p$ where ℓ_c is σ-additive and ℓ_p is purely finitely additive. *Yosida & Hewitt, Finitely additive measures, Trans. AMS (1952).*

2.2 What I proved (turned out classical)

Collective exhaustion (CE) in a directed system: defined as $\exists j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. Under compatibility, the existential witness is vacuous: $\ell_j(\pi_{ij}^{-1}(E_n)) = \ell_i(E_n)$ for all $j \geq i$. So CE $= \sigma$-additivity of each ℓ_i. This is not a theorem; it is the observation that the definition contains a redundant quantifier.

The directed-system extension theorem (Theorem **??** in Paper I) is Kolmogorov extension in directed-system language with "sequential common refinements" replacing the usual product structure.

The Lean formalization (0 sorry on the Carathéodory route, 1 sorry for Yosida–Hewitt on the Stone route) is genuine verified infrastructure, even though the theorems are classical.

2.3 What I understand from this

1. The passage from finite additivity to σ-additivity is the critical non-finitary step in probability. Compatibility, normalization, and finite additivity are all first-order / finitary conditions.

2. No first-order condition can force σ-additivity. *Immediate from Łoś's theorem: the ultraproduct of Dirac masses on $\mathbb{N}$ is a purely finitely additive charge. Hoover (1978), Fagin–Halpern–Megiddo (1990) for the probability-logic formulation.*

3. The finite-cofinite charge on $\mathbb{N}$ is the canonical obstruction: normalized, finitely additive, not σ-additive, and arises as the ultraproduct witness.

4. The Yosida–Hewitt decomposition gives the exact characterization: CE $\Leftrightarrow \ell_p = 0$.

3 Layer 2: From dual space to realization

3.1 The classical results (all known)

- **Stone representation:** Every Boolean algebra is isomorphic to the clopen algebra of a compact, Hausdorff, totally disconnected space (its Stone space). *Stone (1936); Johnstone, Stone Spaces (1982).*

- **Charge $\leftrightarrow$ measure on Stone space:** Every finitely additive charge on a Boolean algebra B corresponds to a regular Borel (Baire) measure on $\mathrm{St}(B)$. *Fremlin, Measure Theory, Vol. 3, §326 and Vol. 4, §416; Halmos (1950).*

- **σ-additivity $\leftrightarrow$ support:** A charge on B is σ-additive iff the corresponding Stone-space measure vanishes on meagre Baire sets, equivalently assigns full mass to countably complete ultrafilters. *Rao & Rao, Theory of Charges (1983), Theorem 10.5.3.*

- **Principal ultrafilters:** Every principal ultrafilter is countably complete. The converse holds for $\mathcal{P}(\kappa)$ when κ is below the first measurable cardinal (Ulam's theorem). For non-σ-complete algebras (typical cylinder algebras), countably-complete non-principal ultrafilters can exist without large cardinals.

3.2 What I proved (turned out classical)

Stone measure exists unconditionally: Any compatible charges produce a Baire probability measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$. This is Stone representation + Choksi. Lean-verified (0 sorry).

Descent requires $\hat{\mu}(\mathrm{pure}(\Omega)) = 1$: concentration on principal ultrafilters. Under discriminability and the assumption that all countably-complete ultrafilters in the relevant subalgebra are principal (which holds for σ-complete algebras below the first measurable cardinal), this is equivalent to σ-additivity. The support characterization is Rao–Rao 10.5.3; the principal-vs-countably-complete gap is the subtlety.

Realization is unconstrained: Any surjective-projecting $S \subseteq \mathrm{St}(\mathcal{C})$ can serve as Ω. This is the abstractness of Boolean algebras recognized since Stone (1936).

3.3 What I understand from this

1. The Stone construction gives the *unconditional* object: $\hat{\mu}$ always exists. It is the "empirically adequate" object in van Fraassen's sense.

2. Descent to Ω is a *choice*: which ultrafilters count as "realized" is additional structure over the algebra.

3. The three-layer nesting:

$$\text{pure}(\Omega) \subseteq \{\text{countably complete ultrafilters}\} \subseteq \text{St}(\mathcal{C}).$$

σ-additivity gives mass on the middle set (intrinsic). Descent needs mass on the inner set (depends on Ω).

4. The van Fraassen identification (Stone measure = empirical adequacy (EA), descent = realist commitment) has no precedent in the literature. It is a philosophical contribution, not a mathematical one.

4 Layer 3: From probability to value-definiteness

4.1 The classical results (all known)

- **Boolean algebras admit 2-valued homomorphisms:** Ultrafilters on Boolean algebras are exactly the 2-valued homomorphisms. Existence by Zorn. *Stone (1936); every Boolean algebra textbook.*
- **Kochen–Specker:** For $L(H)$, $\dim H \geq 3$, no dispersion-free state (2-valued homomorphism) exists. *Kochen & Specker (1967).*
- **Gleason:** For $L(H)$, $\dim H \geq 3$, every σ-additive state is $s(P) = \text{tr}(\rho P)$. *Gleason (1957).*
- **Bub–Clifton:** Given a state ρ and a preferred observable R, there is a unique maximal Boolean subalgebra $D(\rho, R) \subseteq L(H)$ on which the state is a mixture of dispersion-free states. *Bub & Clifton (1996); Halvorson & Clifton (1999).*
- **Orthomodular lattice (OML) duality:** The category of OMLs is dually equivalent to a category of orthomodular spaces. Principal filters $P(A)$ are part of the dual structure (unlike Stone duality where they are invisible). *McDonald & Bimbó (2023), MLQ. Zero citations in quantum foundations as of 2026.*
- **Colimit generation:** Gluing Boolean algebras via categorical pushout produces orthomodular lattices, generically non-distributive. *Gunji, Haruna & Uragami (2026), arXiv:2603.22353.*

4.2 What I proved (classical + one novel framing)

Commensurability theorem: Distributivity is sufficient for value-definite realism (VDR); for $L(H)$, $\dim H \geq 3$, non-distributivity blocks it. Lean-verified (0 sorry, 1 axiom for Kochen–Specker (KS)). *The mathematics is the assembly of Stone + KS + Gleason + Bub–Clifton. Not a new theorem.*

The EA / probabilistic realism (PR) / VDR vocabulary connecting van Fraassen's empirical adequacy to quantum logic via McDonald–Bimbó duality is novel framing. No precedent found.

The coherence filtration $\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{\text{df}}(A)$ with three named transitions (pasting, regularity, sharpness) organizes known results. The chain itself is textbook (Pták–Pulmannová, Kalmbach); the transition names and the observation that distributivity collapses the filtration are new packaging.

4.3 What I understand from this

1. **Distributivity controls value-definiteness.** Boolean algebras have ultrafilters (VDR available). Non-Boolean OMLs (dim ≥ 3) do not (KS). The dividing line is distributivity — sufficient but not necessary in full generality (dim 2, horizontal sums).

2. **The algebra can constrain realization in the OML case.** Unlike Boolean Stone duality, the McDonald–Bimbó dual carries $P(A)$ as structure. "Realization is additional structure" in the Boolean case; it is *constrained* structure in the OML case.

3. **The realism debate has different mathematical answers for different algebras.** Boolean: EA, PR, VDR all available; philosophical choice. OML (dim ≥ 3): VDR blocked; PR available (Gleason); EA always available.

4. **The topos programme is more powerful.** Isham–Butterfield / Döring–Isham / Heunen–Landsman–Spitters encode the same structure via spectral presheaves and Bohrification. Our approach is coarser (single dual space, not presheaf over contexts) but connects more directly to van Fraassen.

5 Delay reconstruction

5.1 The classical results

- **Takens (1981):** Generic smooth observable gives an injective delay map for $L \geq 2 \dim X$.
- **Sauer–Yorke–Casdagli (1991):** Prevalence-based embedding theorem.
- **Generating partitions (Sinai 1959, Krieger 1970):** $h(T, P) = h(T)$ iff P generates.
- **Grassberger–Procaccia:** Correlation integral estimates D_2 and K_2 (Rényi-2 dimension and entropy) from delay data.
- **False nearest neighbours (FNN) (Kennel–Brown–Abarbanel 1992):** Embedding dimension selection. Sensitive to noise (Rhodes–Morari 1997).
- **Ragwitz–Kantz (2002):** Local prediction error minimization for (d, L) selection.
- **Botvinick-Greenhouse et al. (2025):** Measure-theoretic Takens via optimal transport. Lifts embedding to probability spaces.

5.2 What I proved (mostly classical)

Reconstruction theorem: Observable σ-algebra = full σ-algebra mod μ iff delay algebra dense in L^2 iff delay map is measure-theoretic embedding. *This is Krieger's generating partition theorem in delay-map language.*

Predictive factorization, Chapman–Kolmogorov, Koopman–Perron: All classical (Rokhlin, Kolmogorov, Koopman 1931).

Finite-sample convergence rates: Minimax rates $n^{-s/(2s+d)}$ via Stone (1982); elbow stopping via Lepski (1991). *Not cited in the original work — a significant gap.*

5.3 What is genuinely novel (but motivation unclear)

Fibre mixing: A spatial non-degeneracy condition on the Rokhlin disintegration over delay fibres. Not equivalent to any standard temporal mixing condition. Closest relative: Furstenberg relative weak mixing (different condition). *Novel definition. Open derivability question.*

Bridge theorem: Algebraic reconstruction ($\delta \to 0$) $\Leftrightarrow$ geometric reconstruction (collision probability $\to 0$) under fibre mixing. *Novel. But: on standard Borel spaces with fixed L, the two conditions are essentially equivalent anyway. The theorem is about convergence rates as L grows. Motivation unclear — needs an example where the two modes actually diverge.*

Entropy characterization: $\delta(L) \to 0$ iff $H_2(\nu_L) \to \infty$. *Novel. But: a reinterpretation of what Grassberger–Procaccia already computes, not a new tool.*

5.4 What I understand from this

1. The reconstruction problem is *algebraically underdetermined*: the observation algebra does not constrain the realization. Embedding selection (d, L) is necessarily empirical.

2. Every practical reconstruction criterion (FNN, mutual information, Lyapunov convergence, prediction error) implicitly assumes a modelling commitment. Deterministic criteria fail under noise because noise violates the commitment.

3. The Rokhlin distance $\delta(\mathcal{O})$ is the measure-theoretic notion of "how close is the observable σ-algebra to generating everything." It is the Rokhlin metric, named and studied since 1967.

4. Collision probability / Rényi-2 entropy measures the "collision structure" of the fibre decomposition. Grassberger–Procaccia already extracts this from data.

5. Fibre mixing is about the *transition regime*: fibres shrinking but not yet singletons. It is vacuous at both endpoints (no reconstruction, and full reconstruction).

6 Dead ends and lessons

1. **De Finetti via Stone:** Exchangeability forces σ-additivity (Hausdorff moments). The "exotic part" of the Stone measure is empty. Dead end.

2. **Stationary Stone:** The Yosida–Hewitt decomposition commutes with shift-invariance. An observation, not a theorem. The real open question (ergodic theory of the purely finitely additive part) is out of reach.

3. **Dynamics from observation:** "Can the Stone space give rise to dynamics without positing T?" No: the algebra doesn't constrain Ω, so it certainly doesn't constrain structure on Ω.

4. **Paper III (noise thresholds):** Residual autocorrelation as embedding diagnostic = Billings–Voon (1986) / Casdagli (1991). FNN noise sensitivity = Rhodes–Morari (1997). Elbow stopping = Lepski (1991). Rediscovery throughout.

5. **"No intermediate condition":** The claim that no condition interpolates between first-order and σ-additivity is false. $\limsup \ell(A_n) < c$ trivially interpolates.

7 What remains genuinely open

1. **Fibre mixing derivability:** Does ergodicity force ν-a.e. large fibre to be balanced for near-maximizers of the Rokhlin distance? *Well-posed, partially solved (positive fraction proved), but the paper built around it may lack motivation. Need a motivating example first.*

2. **OML extension problem:** Does every state on a general OML extend to a σ-additive measure on the McDonald–Bimbó dual $S_0(A)$? Gleason answers for $L(H)$; the general case is open. *Needs OML expertise to pursue.*

3. **Ergodic theory of purely finitely additive charges:** What are the extreme points of the purely finitely additive shift-invariant charges? Connected to Banach limits. *Out of reach currently.*

8 The understanding, condensed

One sentence per insight:

1. The passage from coherent charges to probability requires σ-additivity, which is not derivable from finitary conditions.
2. The Stone construction gives an unconditional measure on the dual space; descent to a realization requires a commitment the algebra cannot force.
3. Which ultrafilters are "realized" is additional structure in the Boolean case and constrained structure in the OML case.
4. Distributivity is sufficient for value-definiteness; non-distributivity blocks it for $L(H)$, $\dim \geq 3$, but not for all non-distributive OMLs (e.g. $L(\mathbb{C}^2)$, horizontal sums).
5. The realism/empiricism boundary is algebraic: van Fraassen's empirical adequacy is the unconditional level; value-definite realism requires the algebra to admit dispersion-free states (guaranteed by distributivity, blocked by KS for $L(H)$, $\dim \geq 3$).
6. Delay reconstruction is algebraically underdetermined: embedding selection is necessarily empirical.
7. Fibre mixing is a novel spatial condition on the disintegration, distinct from temporal mixing, relevant only in the transition regime.
8. The bridge between algebraic and geometric reconstruction may be solving a problem that doesn't arise naturally.
9. Every practical reconstruction diagnostic implicitly assumes a modelling commitment; noise violates deterministic commitments.
10. Audit before you draft.

9 Key references

Measure extension and charges

- Halmos, *Measure Theory* (1950) — §13, §53–54
- Billingsley, *Probability and Measure* — Ch. 36
- Bogachev, *Measure Theory* — Vol. 1, Ch. 1
- Rao & Rao, *Theory of Charges* (1983) — Thm 10.5.3
- Yosida & Hewitt (1952) — Trans. AMS
- Choksi (1958) — Proc. London Math. Soc.
- Fremlin, *Measure Theory* — Vol. 3 §326, Vol. 4 §416

Stone duality

- Johnstone, *Stone Spaces* (1982)
- Stone (1936) — Trans. AMS

Quantum logic and contextuality

- Kalmbach, *Orthomodular Lattices* (1983)
- Pták & Pulmannová, *Orthomodular Structures as Quantum Logics* (1991)
- Kochen & Specker (1967) — J. Math. Mech.
- Gleason (1957) — J. Math. Mech.
- Bub & Clifton (1996) — SHPMP
- Halvorson & Clifton (1999) — IJTP
- Abramsky & Brandenburger (2011) — New J. Phys.
- McDonald & Bimbó (2023) — MLQ
- Gunji, Haruna & Uragami (2026) — arXiv:2603.22353

Topos approaches

- Isham & Butterfield (1998) — IJTP
- Döring & Isham (2008) — JMP
- Heunen, Landsman & Spitters (2009) — CMP

Delay embedding and reconstruction

- Takens (1981) — Lecture Notes in Math.
- Sauer, Yorke & Casdagli (1991) — J. Stat. Phys.
- Kennel, Brown & Abarbanel (1992) — Phys. Rev. A
- Fraser & Swinney (1986) — Phys. Rev. A
- Ragwitz & Kantz (2002) — Phys. Rev. E
- Botvinick-Greenhouse et al. (2025) — J. Stat. Phys.
- Kantz & Schreiber, *Nonlinear Time Series Analysis* (2004)

Ergodic theory

- Rokhlin (1949/1962) — disintegration
- Rokhlin (1967) — entropy lectures, Rokhlin distance
- Sinai (1959) — generating partitions
- Krieger (1970) — generator theorem
- Furstenberg (1977) — relative mixing, Szemerédi
- Shields, *Ergodic Theory of Discrete Sample Paths*

Philosophy of science

- van Fraassen, *The Scientific Image* (1980)
- van Fraassen, *Quantum Mechanics: An Empiricist View* (1991)
- de Finetti, *Theory of Probability* (1974)
- Pitowsky, *Quantum Probability — Quantum Logic* (1989)

Nonparametric statistics

- Stone (1982) — minimax rates
- Lepski (1991) — adaptive estimation
- Hoeffding (1948) — U-statistics
- Billings & Voon (1986) — NARMAX residual validation
- Casdagli (1991/1992) — deterministic vs stochastic

Model theory and probability logic

- Łoś (1955) — ultraproduct theorem
- Hoover (1978) — probability logic
- Fagin, Halpern & Megiddo (1990) — probability logic